

CATHOLIC JUNIOR COLLEGE  
General Certificate of Education Advanced Level  
Higher 2

## MATHEMATICS

Paper 1

**9740/01**

**3 September 2009**

**3 hours**

Additional Materials:    Answer Paper  
                                  Graph Paper  
                                  List of Formulae (MF 15)

### READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams or graphs.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

**At the end of the examination, arrange your answers in NUMERICAL ORDER. Place this cover sheet in front and fasten all your work securely together.**

The number of marks is given in brackets [ ] at the end of each question or part question.

| Question No. | Marks | Question No. | Marks        |
|--------------|-------|--------------|--------------|
| 1            | / 4   | 7            | / 10         |
| 2            | / 6   | 8            | / 11         |
| 3            | / 6   | 9            | / 11         |
| 4            | / 7   | 10           | / 14         |
| 5            | / 7   | 11           | / 14         |
| 6            | / 10  | <b>TOTAL</b> | <b>/ 100</b> |

This document consists of **5** printed pages.

[Turn Over]

1 Solve the inequality  $x^2 - 3x - 4 < 0$ . [1]

Hence find the exact value of  $\int_{-1}^{3\sqrt{2}} |x^2 - 3x - 4| dx$ . [3]

2 Referred to the origin  $O$ , the position vectors of the points  $A$  and  $B$  are  $-3\mathbf{i} + 9\mathbf{k}$  and  $3\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$  respectively.

(i) Find the position vector of the point  $M$  on the line segment  $AB$  such that  $AM : AB = 2 : 3$ . [2]

(ii) Show that  $OM$  is perpendicular to  $AB$ . [2]

(iii) The point  $C$  is the reflection of the origin  $O$  in the line  $AB$ . Calculate the area of triangle  $OAC$ . [2]

3 Find

(i)  $\int x \sin x dx$ , [2]

(ii)  $\int \frac{1}{x\sqrt{1-x}} dx$ , where  $0 < x < 1$ , using the substitution  $x = \cos^2 u$ . [4]

4 The  $r$ th term of a sequence is given by  $u_r = r(3r + 1)$ ,  $r = 1, 2, 3, \dots$

(i) Write down the values of  $\sum_{r=1}^n u_r$  for  $n = 1, 2, 3$ , and  $4$ . [1]

(ii) Make a conjecture for a formula for  $\sum_{r=1}^n u_r$ , giving your answer in the form  $nf(n)$ , where  $f(n)$  is a function of  $n$ . [1]

(iii) Prove by induction a formula for  $\sum_{r=1}^n u_r$ . [5]

5 Use the substitution  $y = \frac{u}{x}$  to show that the differential equation

$$\frac{dy}{dx} = \frac{2xy - 2y - 1}{x^2 + 2x} \text{ reduces to the differential equation } \frac{du}{dx} = \frac{3u - 1}{x + 2}.$$

Hence find the general solution for  $y$  in terms of  $x$ . [7]

- 6 Given that  $y = e^{(1+x^2)}$ , prove that

$$\frac{d^2 y}{dx^2} = 2 \left( y + x \frac{dy}{dx} \right). \quad [2]$$

Hence expand  $e^{(1+x^2)}$  in ascending powers of  $x$  up to and including the term in  $x^4$ . [6]

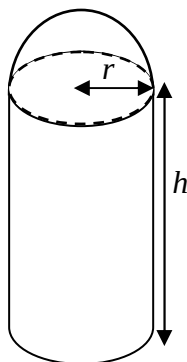
Use this series expansion to estimate the value of

$$\int_0^{\frac{1}{2}} e^{\left(\frac{4+x^2}{4}\right)} dx, \text{ correct to 3 decimal places.} \quad [2]$$

- 7 A candy maker is interested in using containers in the shape shown below to package his candies. The container is made up of an open cylinder of height  $h$  cm and radius  $r$  cm, with a hollow hemispherical lid of radius  $r$  cm.

In order to minimise production cost in this difficult time, the candy maker wants to use containers with the least surface area while maintaining the volume of each container at  $500 \text{ cm}^3$ . If the material used to construct the container cost \$0.015 per  $\text{cm}^2$ , find, using differentiation, how much a container with minimum surface area costs to the candy maker. Leave your answer to 2 decimal places. [10]

[Volume of sphere,  $V = \frac{4}{3}\pi r^3$  ; Surface area of sphere,  $S = 4\pi r^2$ ]



- 8(a) One of the roots of the equation  $z^4 - z^3 + 4z^2 + 3z + p = 0$  is  $1 - 2i$ , where  $p$  is a constant.

A student says that “One of the other roots must be  $1 + 2i$ .”

Explain why this statement is not entirely correct. [1]

(i) Determine the value of  $p$ . [1]

(ii) Find the exact values of all the other roots of the equation. [3]

- 8(b) The fixed complex number  $a$  has modulus  $R$  and argument  $\alpha$ , and  $a^*$  denotes the conjugate of  $a$ . Given that  $q = \frac{a^3}{a^*}$ , express  $q$  in trigonometric form. [3]

If  $q^{\frac{1}{6}}$  is purely imaginary, determine the least positive value of  $\alpha$ . [3]

- 9 A curve is defined by the parametric equations  $x = 4t^2$ ,  $y = \frac{2t}{t^2 + a}$ , where  $a$  is a positive constant.

(i) Using differentiation, find the turning points of this curve in terms of  $a$ . [4]

(ii) Show that the cartesian equation of the curve is  $y^2 = \frac{16x}{(x + 4a)^2}$ . [2]

(iii) Sketch  $y^2 = \frac{16x}{(x + 4a)^2}$ , indicating clearly the coordinates of the points where the graph crosses the axes, turning points and the equations of any asymptotes. [3]

(iv) Describe a sequence of geometrical transformations that map the graph of  $y^2 = \frac{16x}{(x + 4a)^2}$  onto the graph of  $y^2 = \frac{16(3x - 4a)}{9x^2}$ . [2]

- 10(a) Susan baked 2500 cookies. She decided to give part of them to her friends and sell the remainder for charity.

(i) To pack the cookies meant for her friends, Susan placed 5 cookies in the first bag. Each subsequent bag she packed contained double the number of cookies in the previous bag. How many complete bags of cookies did Susan have to offer her friends? [3]

(ii) To pack the remaining cookies meant for sale, Susan placed 4 cookies in the first bag. Each subsequent bag she packed contained 3 cookies more than the previous bag. If she charged \$0.50 for each cookie, how much would the last complete bag of cookies cost? [5]

- (b) Given that  $T_r = 2^{2+rb}$ , where  $b$  is a constant,

(i) show that the terms of the series  $\sum_{r=1}^n \ln T_r$  form an arithmetic progression, [2]

(ii) express  $S_n$ , sum of the first  $n$  terms of this arithmetic progression, in terms of  $n$  and  $b$ , [2]

(iii) hence find an inequality satisfied by the constant  $b$  such that the difference between  $S_{13}$  and  $S_{14}$  is not more than 0.5. [2]

11 The planes  $p_1$  and  $p_2$ , which meet in the line  $l$ , have vector equations

$$\mathbf{r} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mu_1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \mu_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

respectively, where  $\lambda_1, \lambda_2, \mu_1$  and  $\mu_2$  are real constants.

- (i) Show that  $l$  is parallel to the vector  $5\mathbf{i} + 6\mathbf{j} + \mathbf{k}$ . [3]
- (ii) Calculate the acute angle between  $p_1$  and  $p_2$ . [2]
- (iii) Find, in exact form, the perpendicular distance from the point with coordinates  $(4, 2, 2)$  to  $p_2$ . [2]

The plane  $p_3$  has equation  $ax - 2y + 2z = b$ , where  $a, b \in \mathfrak{R}$ .

- (iv) Find  $b$  in terms of  $a$  such that all three planes meet at the single common

point with position vector  $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ . [4]

- (v) If given instead that  $a = 2$ , find the values of  $b$ , such that the distance between the planes  $p_1$  and  $p_3$  is  $\frac{1}{\sqrt{3}}$  units. [3]